

PREDICTING APARTMENT RENTAL PRICES IN SWITZERLAND

DSPRO1 Project

Team:

Elias Martinelli, Timo Schlumpf

Supervisor:

Elena Nazarenko



AGENDA

1

The Problem

2

Scope Status

3

Data Source & Enrichment

4

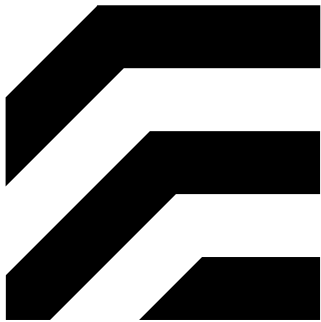
Baseline Result

5

Next steps

6

Questions & Feedback



1

THE PROBLEM

THE PROBLEM

You find a listing
3 rooms, 80 m², Zürich

CHF 2 400 / mo
Is this fair?

Why it's hard to judge

Location dependency
Same flat costs 2× more across cantons

Many interacting features
Area, floor, year built, amenities ...

No transparent benchmark
No public reference model exists

→ **We build a supervised ML model that predicts fair rent from listing data**

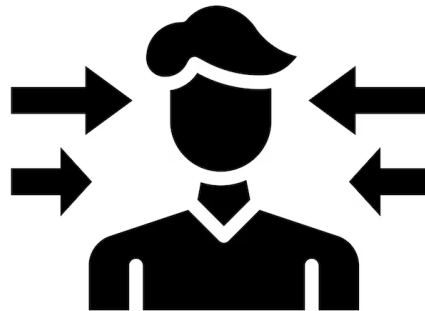
WHY IS IT RELEVANT?



- Tenants



- Landlords



- Socially

HYPOTHESIS

*Location-related variables combined with structural apartment features contain sufficient signal to predict monthly rental prices in Switzerland with practically useful accuracy and a **Random Forest Regressor** will outperform a Linear Regression model in doing so.*

2

SCOPE STATUS

SCOPE

- **In Scope**

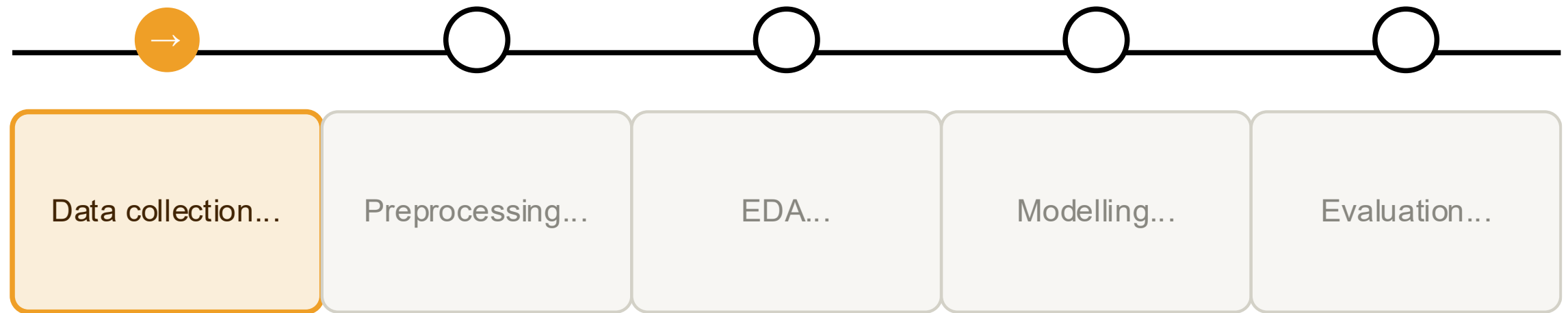
- ✓ Web scraping of Swiss rental listings
- ✓ Data cleaning, deduplication & outlier removal
- ✓ Exploratory Data Analysis (EDA)
- ✓ Feature engineering (location, structure, amenities)
- ✓ Baseline model → Linear Regression → Random Forest
- ✓ Model evaluation: MAE, RMSE, R^2
- ✓ Experiment tracking with MLflow
- ✓ Version control & Kanban via GitHub Projects

- **Out of Scope**

- Cloud deployment or production-ready application
- Real-time or live price feed integration
- Time-series / rental price forecasting
- Non-apartment property types (houses, commercial)
- User-facing UI or web app

STATUS

Project Progression



 Upcoming

 In progress

 Upcoming

PROJECT MANAGEMENT TOOL



- Backlog
- Roadmap
- Milestones

3

DATA SOURCE & ENRICHMENT

DATA SOURCE

Webscrapping Data from



comparis.ch

Wohnung mieten in der Schweiz: 27'829 Resultate

Favoriten (0)



Alle verfügbaren Wohnungen und Häuser der Schweiz auf comparis.ch. Für Sie gefunden auf comparis.ch, immobilier.ch, properstar.ch, immoscout24.ch, homegate.ch und mehr als 300 Weiteren.



Filter (3)

Suchabo anlegen

Sortieren nach:

Erstmals gefunden am



Wohnung

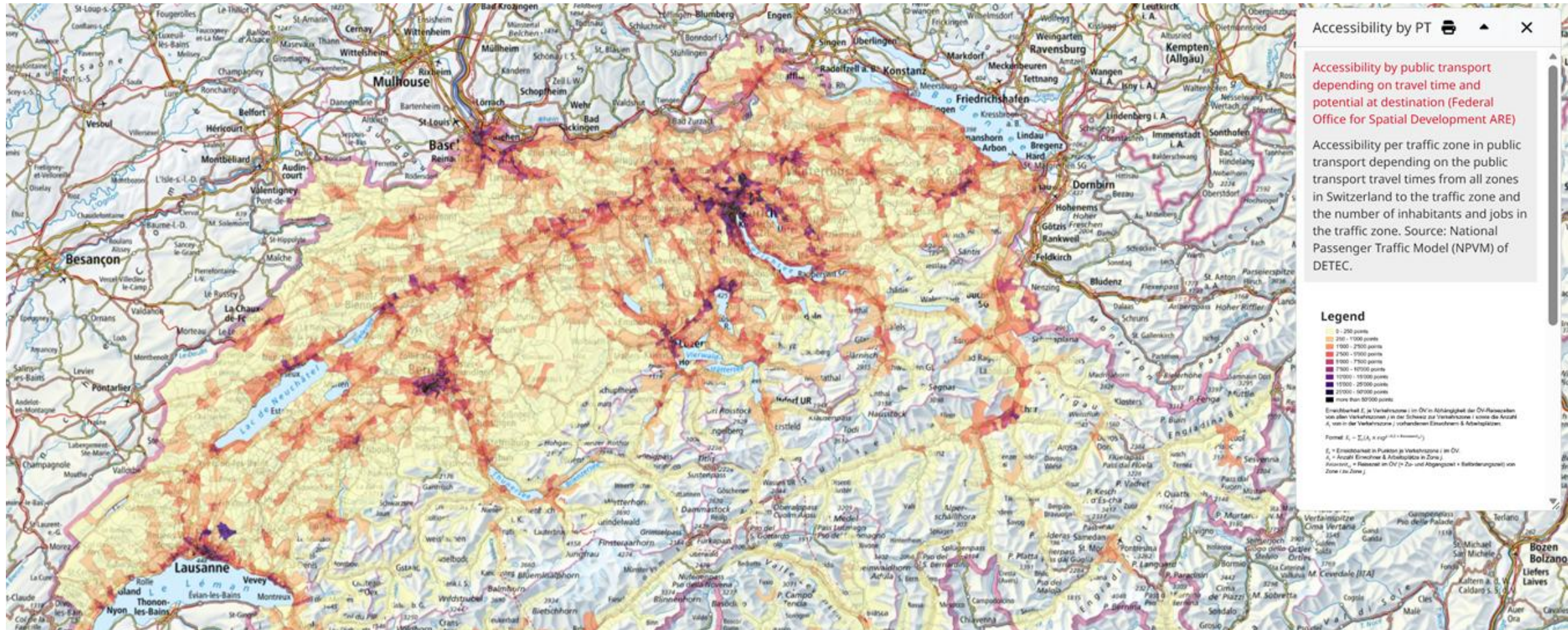
Online seit 32 Minuten

3.5 Zimmer, 76 m², 1. Etage

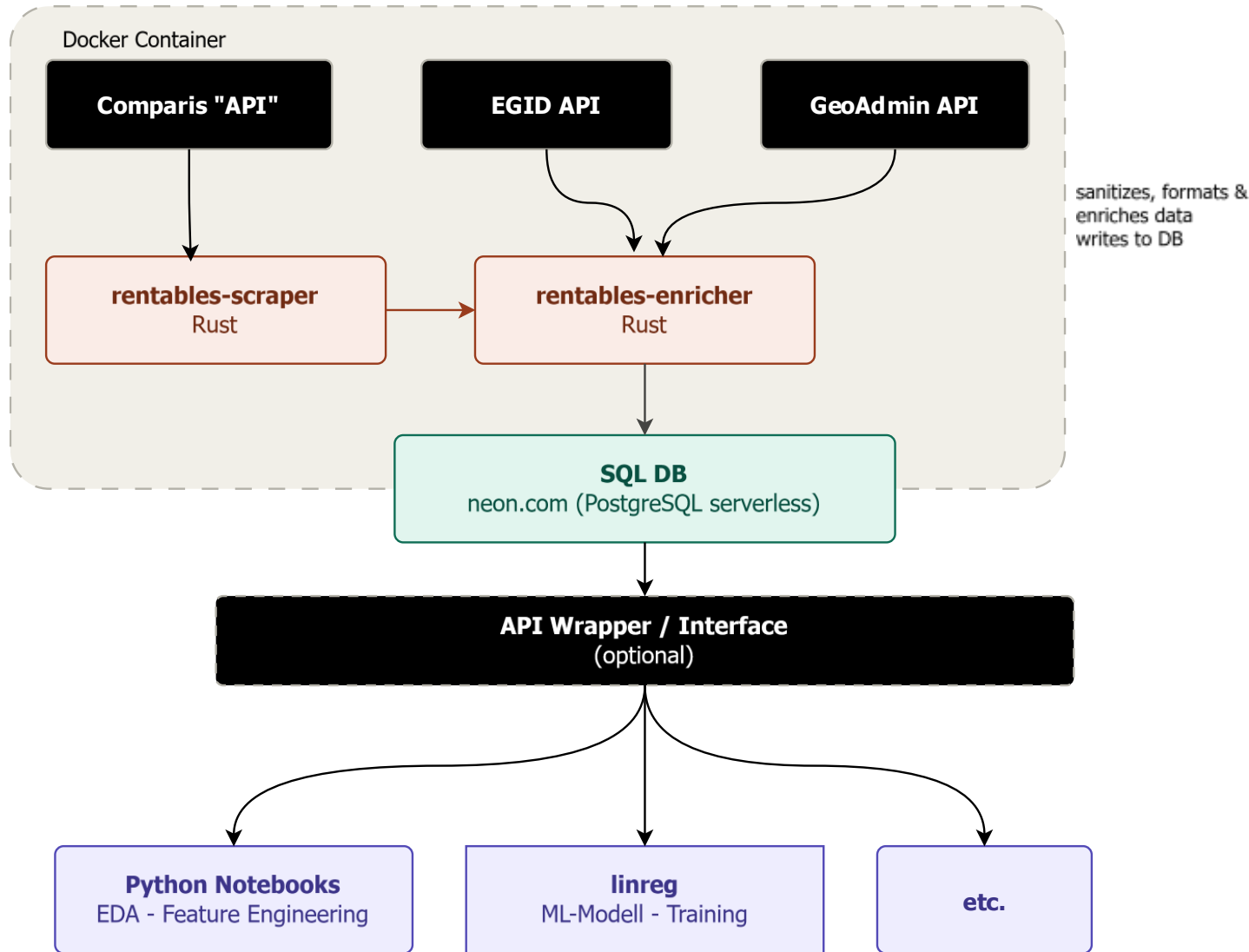
«SUPERBE APPARTEMENT MODERNE»

DATA ENRICHMENT

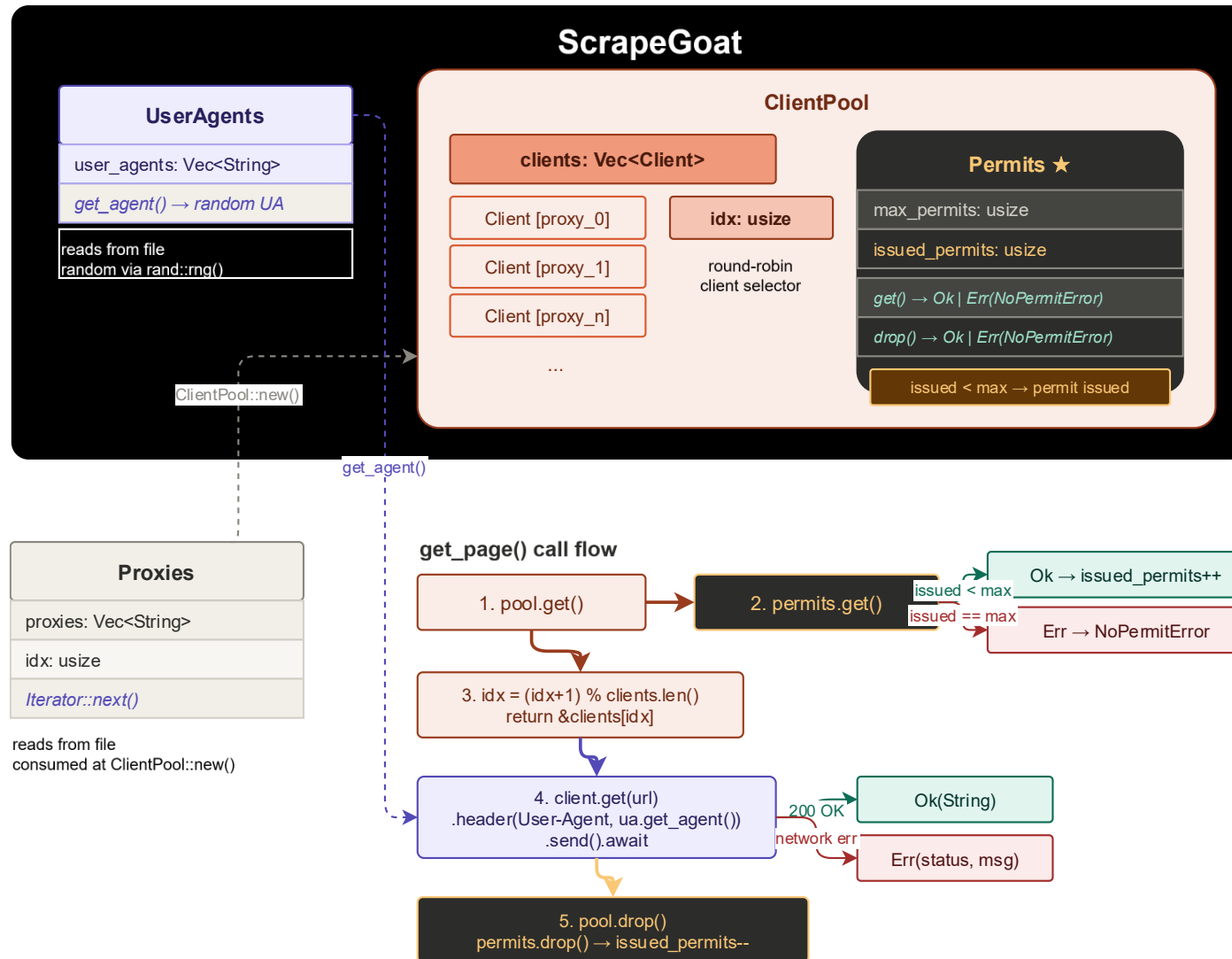
- Over API => geo.admin.ch



DATA PIPELINE HIGH-LEVEL DESIGN



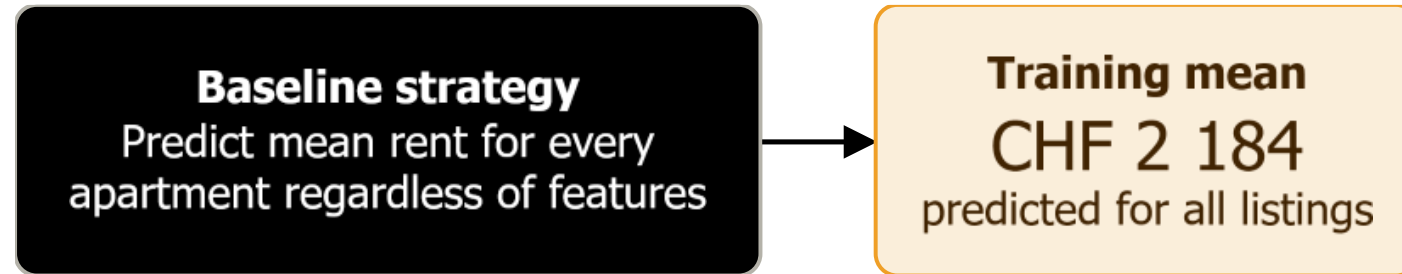
POOL & PERMIT SYSTEM



4

BASLINE RESULTS

BASILINE RESULTS



Baseline performance



Interpretation
Off by CHF 612 on average — this is our lower bar. Any real model must clearly beat this.

5

NEXT STEPS

FROM LISTING TO PREDICTION



MODEL COMPARISON STRATEGY



Input features

Location



Structural



Amenities



Target variable

Monthly rent in CHF (continuous)

THANK YOU FOR YOUR ATTENTION.

We are happy to answer any questions or discuss the presented topics further.

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Luzern

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